

# HAO PAN

AI Application Engineer | Data Agent | LLM Applications

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## Summary

Master's candidate in Applied Artificial Intelligence with experience in BI analytics, AI application development, and technical teaching. Skilled at translating business problems, metric definitions, and acceptance criteria into testable Agent workflows, then delivering end-to-end prototypes spanning frontend, backend, evaluation, and iteration. Prioritizes measurable time savings, reviewable outputs, and safe failure behavior.

## Education

### Nanyang Technological University

Nov. 2025 – Expected Jan. 2027

Master of Computing in Applied Artificial Intelligence, College of Computing and Data Science

Singapore

### Suzhou University

Sep. 2019 – Jun. 2023

B.Eng. in Data Science and Big Data Technology; China National Scholarship (Top 0.2% nationwide)

China

## Experience

### Shopee Singapore

Mar. 2026 – Present (through Sep. 2026)

Data Analyst Intern (AI Focus), Buyer & Marketing BI Team

Singapore

#### A/B Analysis Agent Skill Development & Reliability Evaluation

- Partnered with product managers and regional stakeholders to align inputs, metric definitions, and acceptance criteria, then packaged five recurring A/B analysis workflows (PDP, OPC, Review, Cart, and Item) as reusable Agent Skills for team use. Created a common Skill template covering parameters, SQL, output structure, and pre-output validation, reducing turnaround from **30–45 minutes to 2–3 minutes** for supported requests.
- Addressed SQL rewrites, query timeouts, incomplete outputs, and upstream table changes with pinned SQL versions, parameter validation, queries split by experiment group, and completeness checks. Codified **15 failure modes** into regression checks; under matched SQL versions and parameters, all **19 fields across 18 result rows** matched the manual SQL output.

#### Analysis KB: Team Analytics Case Knowledge Base

Internal tool, ongoing

- Led the ongoing development of Analysis KB, a team-wide knowledge base designed to onboard all historical analytics cases. To make fragmented, cross-format materials consistently readable and reusable by AI, designed a unified Case Bundle schema and import workflow that converts Requests, Reports, SQL, and CSV files from Tracking Sheets, Google Docs / Sheets, and individual folders into Markdown, with GitLab as the source of truth.
- Reviewed **160 records** from the previous six months and validated the MVP with the first **5 real cases**, covering 4 Requests, 5 Reports, 12 SQL files, and 9 CSVs. Built the Svelte / Python web app and validated the Search → Compare → Draft workflow, where users confirm candidate cases before Request Draft generation; all 30 generated Markdown files passed automated validation with **0 errors and 0 warnings**.
- Prepared progressive full-case onboarding through an extensible Case Schema, Catalog, and validation pipeline. Pinned each run to a Git commit and recorded evidence paths; backend validation, persisted state, and idempotent handling preserved traceability across refreshes, retries, and restarts, while historical SQL remained view-only.

### Jinken Vocational and Technical College

Mar. 2023 – Oct. 2025

Instructor & AI Technical Lead

China

- Designed and delivered Python, machine learning, and generative AI courses for **300+ students**, turning abstract concepts into labs, assessments, and project-based assignments.
- Led a STEM robotics project and mentored a student team using ROS to decompose, integrate, and iterate an indoor autonomous-flight and obstacle-avoidance solution, winning **Third Prize at the RAICOM 2024 National Finals**.

## AI Projects

### BI Data Agent Sandbox | TypeScript, React, Node.js / Vercel Serverless, AlaSQL, Vitest | [Demo](#) · [GitHub](#)

Jul. 2026

- Built a deterministic BI Data Agent workflow to make SQL generation testable and failures diagnosable. The system maps business questions to intent, metric / schema context, and validated read-only SQL; a Node / Vercel Serverless API executes queries on synthetic e-commerce data and streams result rows, charts, a grounded answer, and an auditable trace over NDJSON.
- Defined request, transport, and terminal-result contracts; used Zod, SQL guardrails, and client-side integrity checks to validate schemas, run identity, event order, trace parity, and terminal state. Blocked sensitive exports before execution, routed unsupported questions to needs-review, and added versioned regression evaluation, contract tests, and CI gates for type checking, tests, linting, and builds.

### ContextCue | TypeScript, Next.js, React | [Demo](#) · [GitHub](#)

Jun. – Jul. 2026

- Built a context layer for personal Agents to prevent cross-person memory contamination, stale recall, and unsafe memory writes. Converted simulated voice commands, chats, and AI feedback into structured memories with person, task, TTL, and source fields, while preserving citations in responses.
- Designed MemoryGate to enforce person and task scope, human confirmation, and TTL before memory writes and reads, while preventing raw chats and audio from entering long-term memory. Created **16 repeatable V0.1 evaluation scenarios** covering context assembly, identity crossover, privacy, expiration, evidence citation, and tone safety.

## Technical Skills

**Languages & Data:** Python, TypeScript, SQL (Presto / Trino, Hive, MySQL), PySpark, SQLite, Shell

**AI & Data Agents:** Google ADK, Agent workflow / Skill design, Text-to-SQL, RAG, LLM evaluation, Guardrails, Langfuse

**Web & Engineering:** React, Next.js, Svelte, Node.js, Vitest, pytest, Git / GitLab, Linux, Vercel, ROS

**Honors:** Huawei ICT Competition Global Final, Cloud Track, Global Third Prize (Top 5%, Jun. 2022)